

1 Robot-Whisper

Neurons see the state the tokens don't.

A Training-Free Embodied Anomaly Alarm from MoE
Routing Dynamics



Observe — Deep MoE routing reveals when the closed loop stalls.



Detect — A self-normalized routing-speed alarm catches failures online.



Act — The alarm can trigger recovery by resampling action noise.



864

rollouts



80%

failure recall



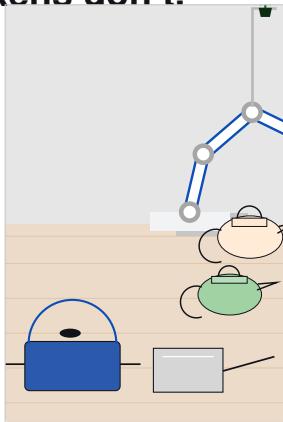
17%

false-alarm rate

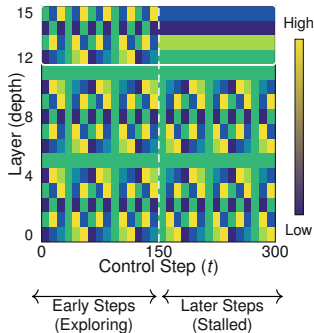


Online

intervention



Deep Routing Pattern
(HB 12–15)



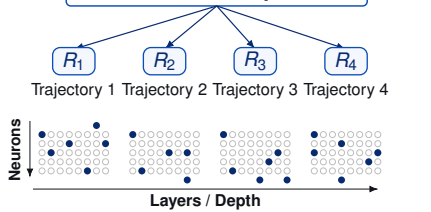
The signal is not who is active. It is whether the pattern is still moving.

2 From Neuron Agreement to Embodied State Sensing

From deciding whether a reasoning trajectory is reliable to sensing whether a robot is still making progress.

NAD (Reasoning LLMs)

Question: What is the capital of France?

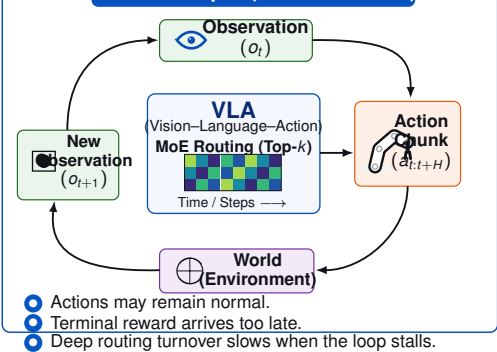


- Outputs may disagree.
- Correct trajectories show stronger neuron agreement.
- Internal activations can guide selection.

From reasoning trajectories to embodied trajectories

If internal activity reveals answer reliability, can internal dynamics reveal when an embodied closed loop has stalled?

Robot-Whisper (Embodied VLA)



NAD (Reasoning LLMs)

• Object	Reasoning trajectory
• Compare across	Samples
• Internal signal	Neuron activations
• Goal	Reliability
• Downstream use	Select

Robot-Whisper (Embodied VLA)

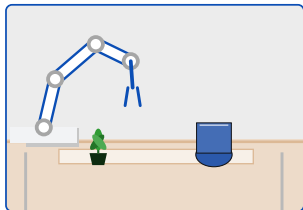
Embodied trajectory
Time
MoE routing sets
Current state
Alarm

NAD compares samples to estimate quality. Robot-Whisper compares time steps to estimate state.

3 Same Model, Same State, Two Different Futures

Only the diffusion sampling noise changes.

Shared simulator state
(after first pot placed)



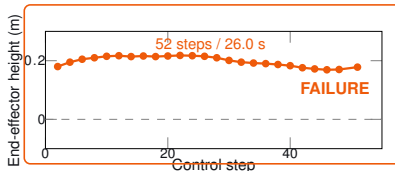
Same model, same state,
same task.



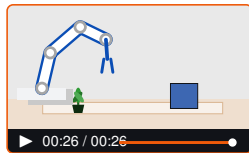
One state, two futures.

The action stream never
emits an explicit error.

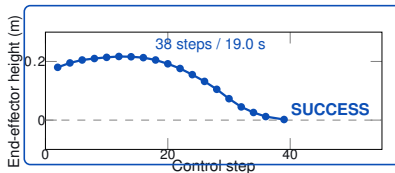
Original noise → failure



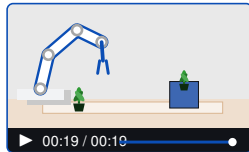
Stalled above second pot



Alternative noise → success



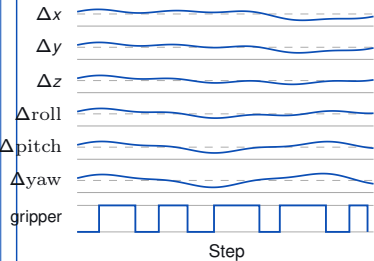
Successful completion



**Closed-loop failure can emerge even when the
model, state, and task are otherwise identical.**

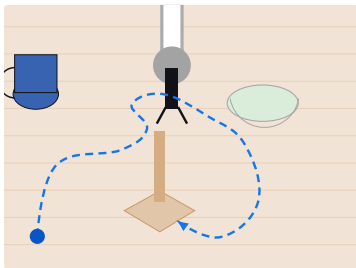
4 Why Can't We Detect the Failure from the Output?

Actions keep coming



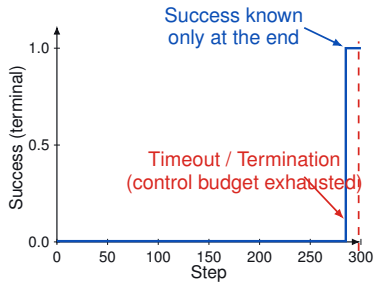
Each control step still outputs a valid action chunk.

Motion is not progress



The robot may still move while task progress has stopped.

Reward arrives too late



Timeout confirms failure only after the control budget is exhausted.

Normal actions

→ continued motion →

late terminal signal

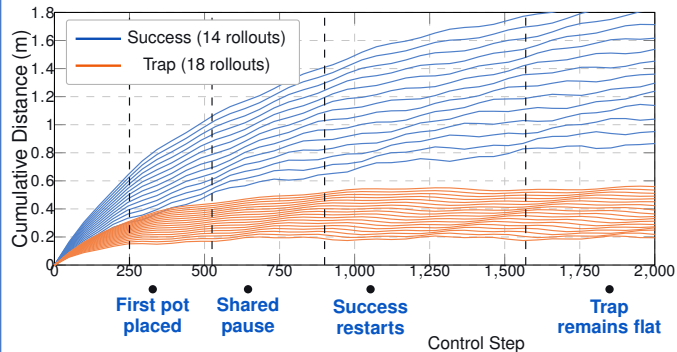


So we turn from what the model says to how its internal state moves.

5 A Trap Is Motion Without Progress

Representative stagnation case

Cumulative Change in End-Effector Position



Statistics / Description

- **32** rollouts from the same state
- **14** successes
- **18** traps
- Only difference: sampling noise

Two trap morphologies

- Stagnation (hovering)
- Looping (repetitive cycles)

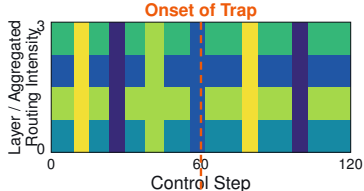
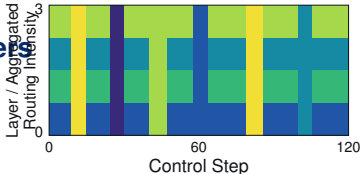
A trap is not the absence of motion. It is the absence of meaningful progress.

6 Deep Routing Freezes When the Closed Loop Stalls

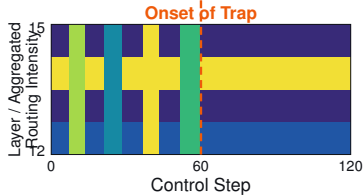
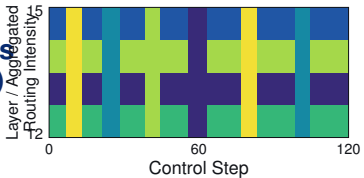
Success Rollout

Trap Rollout

Shallow Layers
(HB 0-3)



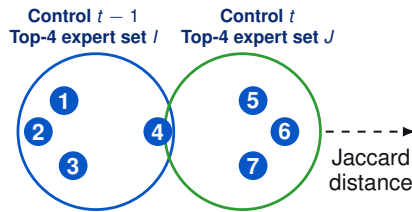
Deep Layers
(HB 12-15)



- The same experts can appear in both results.
- Static identity reflects scene context.
- What matters is whether the pattern keeps changing.

The signal is not expert identity. It is the loss of routing turnover.

7 A Self-Normalized Routing Speed Turns Freeze into an Alarm



Per- (ℓ, u)
Jaccard distance

$$\delta_{t,\ell,u} = 1 - \frac{|I \cap J|}{|I \cup J|}$$

Average across
layers and users

$$d_t = \frac{1}{LU} \sum_{\ell=1}^L \sum_{u=1}^U \delta_{t,\ell,u}$$

Self-normalized
routing speed

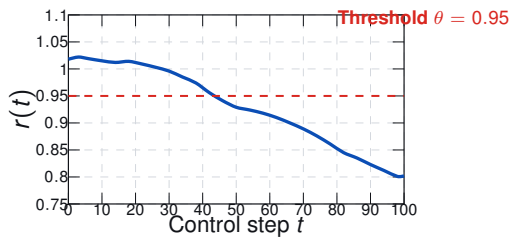
$$r(t) = \frac{\text{mean}(d_{t-W+1:t})}{\text{mean}(d_{1:W_0})}$$

Window sizes: $W = 4, W_0 = 8$

Alarm rule

Alarm if
 $r(t) < \theta$ for
 $K = 3$ steps,
 $\theta = 0.95$

Example $r(t)$



No learned probe

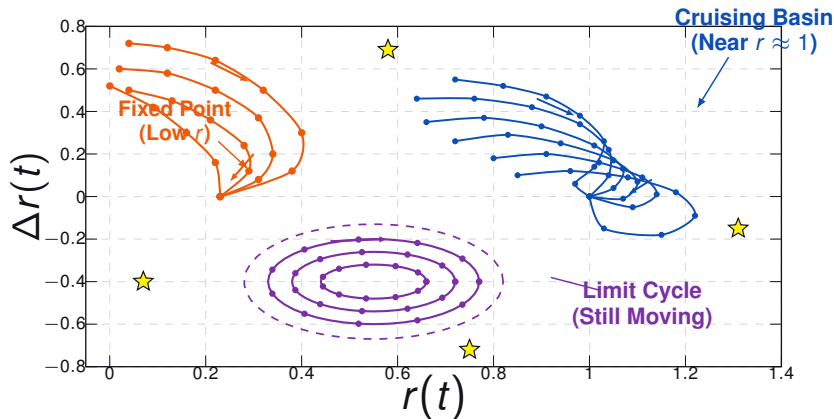


Self-normalized
per rollout



Invariant to
expert renumbering

8 Stagnation Is a Fixed Point; Looping Is a Limit Cycle



- Success Rollouts (Cruising Basin)
- Stagnation Traps (to Fixed Point)
- Looping Traps (Limit Cycle)
- Alarm Point

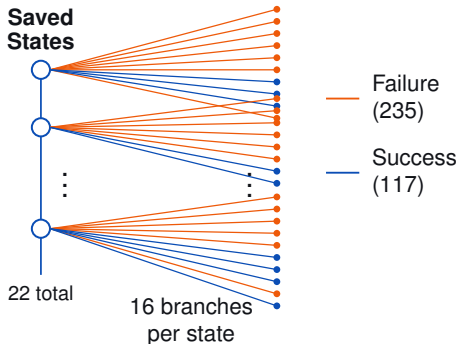
Robot-Whisper is primarily a fixed-point detector, not yet a universal failure detector.

9 One Fixed Rule Detects Stalls Across 352 Controlled Branches

Experiment Summary

- 22 saved states
- 16 noise branches per state
- 352 rollouts total
- 235 failures
- 117 successes

Branch Outcomes (352 rollouts)



Detection Results (Fixed Rule)

- 184 failures caught
- 51 failures missed
- 98 successes passed
- 19 false alarms

Accuracy: 80.1%

Majority baseline: 66.8%

Same rule, fixed threshold, no retraining.

10 Across 864 Rollouts, the Alarm Catches Four Failures out of Five

Two independent batches, same rule, same threshold $\theta = 0.95$

Corpus	Rollouts	Accuracy	Majority Baseline
Archived branches	352	80.1%	66.8%
Independently collected	512	82.0%	57.8%

81.2%

Accuracy

80%

Failure Recall

17%

False-Alarm Rate

+19.7 pts

Over Majority Baseline

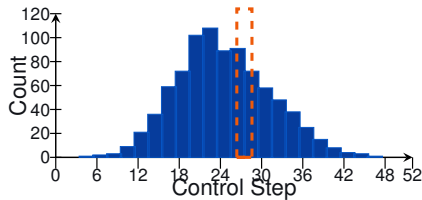


A single training-free rule transfers across independently collected rollout batches.

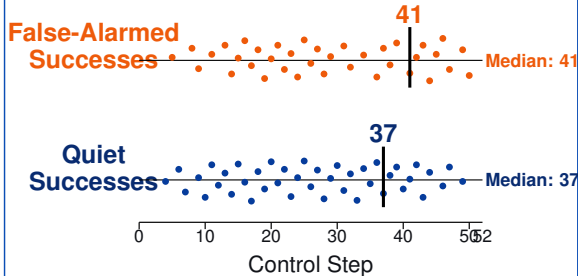
11 The Alarm Arrives Halfway — and Mainly Flags Genuinely Slow Runs

When Does the Alarm Fire? (All Alarms)

Median: q28
(~54% of 52-step
budget)



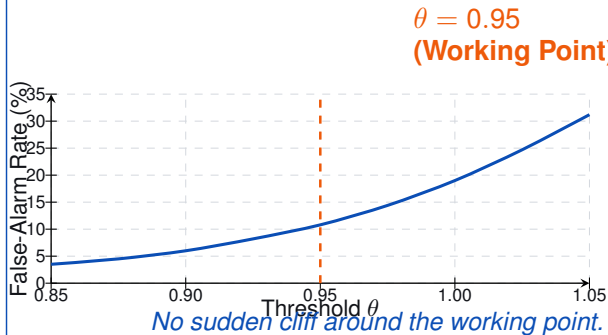
Success Rollout Lengths (Control Steps)



✓ The alarm means 'failing now,' not 'doomed eventually.'

12 The Conclusion Does Not Sit on a Threshold Knife-Edge

False-Alarm Rate vs. Threshold θ



Two Corpora



Same θ



Similar False-Alarm Profile



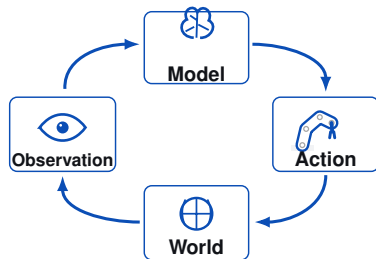
Corpus A (352) / Corpus B (512)



A self-normalized rule remains stable around the chosen threshold.

13 Why It Works: A Compressed Echo of a Stopped World

Closed-Loop System



Evidence

- Routing stickiness $\approx$ input stability: $r \approx 0.93$
- After removing input-stability effect, routing signal drops to near-random.

When the Loop Stalls

- 1 World change weakens
- 2 New observations stay similar
- 3 Router inputs stabilize
- 4 Selected expert sets stabilize



Not Hidden Intention / Not Future Prophecy

The signal appears only after loops stall and disappears when dynamics resume.



A Free Internal Readout / No Extra Forward Pass

The signal is read from existing hidden states—no additional inference or computation.



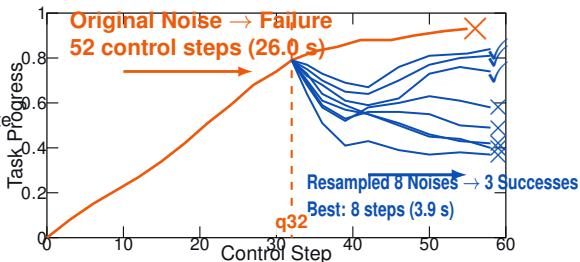
The routing signal is an internal state readout of a stalled closed loop.

14 The Alarm Can Trigger Recovery by Resampling the Action Noise

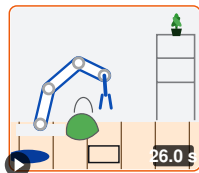
Procedure

- 1  Run a new rollout
- 2  Detector fires online
- 3  q32 alarm triggers
- 4  Save simulator state
- 5  Resample 8 new noise streams
- 6  Compare futures

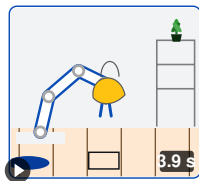
Progress (Task Metric)



Original noise (failure)



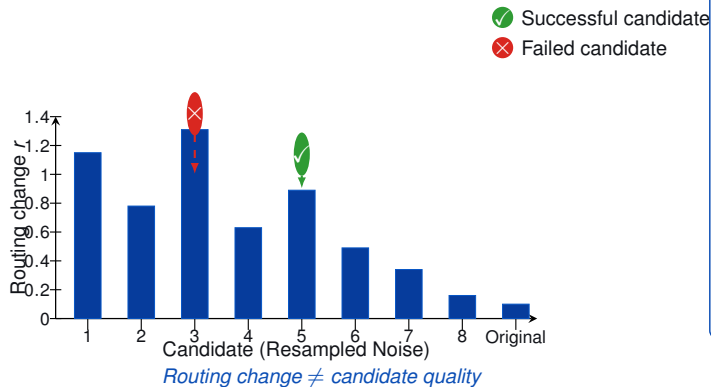
Resampled noise (success)



3 / 8 Succeeded • Best: 8 steps (3.9 s) • Original: 52 steps (26.0 s)

15 Detection Works; Prediction and Recovery Selection Remain Open

Routing Change of 8 Candidates (at Alarm State)



Current Limitations



Recoverability:

Another state $\rightarrow$ 0 / 8 rescued



Trigger optimality:

Random earlier split $\rightarrow$ 3 / 8



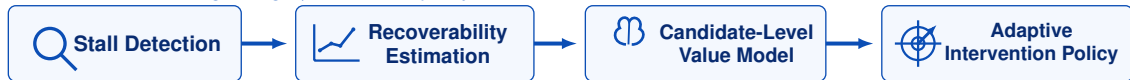
Candidate selection:

Routing change $\neq$ quality



Forecasting:

No stable early warning;
looping traps remain hard



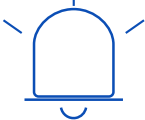
16 The Model Was Already Logging Its State — We Finally Read It

Observe



Deep MoE routing freezes
when the loop stalls,
while actions still look normal.

Detect



A training-free, self-normalized,
expert-invariant online
detector.

Act



An alarm that triggers recovery
by resampling noise —
and exposes true limits.

The routing dynamics reveal what state the closed loop is in.

向内探寻，向外进化。
Seek within. Evolve beyond.

